Using ResearchGPT for learning Active Inference and doing Ant research

Source: YouTube **Playlist**: Active Inference Journal **Duration**: 9:25 | **Uploaded**: Unknown **Transcript method**: auto_caption

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all right the goal of this video is to go through a conversation that I just had with research GPT by siace at types set. to demonstrate some capacities of this model and also learn about active inference ants and Beyond the prompts that I wrote will be read and answers will only be overviewed pause the video or see the sharable conversation Link in the video description for Fuller text so I jumped right in with the prompt provide a comprehensive account of the relationship between active inference and the free energy principle it provided five relevant citations and so I wondered about this in the context of developing a PhD dissertation around the philosophical articulations of active inference and the free energy principle to go beyond just literature search and see if it could output a full dissertation it searched and again returned a list of five papers which was not exactly what I was looking for but they are relevant papers I then pulled into a more beginner or learner line of questioning can you provide a bullet point list of the 31 most important terms for my total comprehensive understanding of active inference it could be technical metaphysical pragmatic or whatever kind of terminology is going to catalyze this Today's starting point of my curiosity and the terms that it recommended were active inference free energy principle basy and brain predictive coding marov blanket variational free energy generative model surprise Precision belief updating epistemic action entropy predictive error perceptual inference hierarchical models action selection propri acception affordances embodied cognition neurophenomenology decision Theory optimal control theory homeostasis allostasis interception empirical base metacognition cognitive dissonance in activism sensory attenuation phenomenal Consciousness I wrote seeing all these terms makes me truly excited however I am not sure where or how to begin based upon evidence supported techniques and learn learning theory can you write me a pseudo code loop that I can follow during the day while I study active inference it provided the pseudo code the study session is initialized and three study sessions occur per day during each study session start with A Brief Review focus on one key term or concept read and understand the concept apply Active Learning techniques reflecting and mind maps relate the New Concept to an already learned concept test understanding take a short break repeat with a new concept or deepen understanding of the same concept and the session reflect and plan take a longer break to help with memory consolidation increment the study session and reflect and plan at the end of the day optional to engage in related discussions or online forums for Community Learning and that is explained excellent already I feel my confidence increasing and veilance rising and I did can you provide citations that address specifically professionally how active inference and reinforcement learning are similar and different several extremely relevant papers including probably the most relevant paper are provided so five relevant papers great work those are some of the right citations for us to consider can you summarize similarities and differences between active inference and reinforcement learning I am looking for a bullet point list that contains as many lines as needed such that every line is composed to be semantically univocal and among lines the entire semantic space of the best possible answer is surveyed thank you and a list was provided with similarities seven of them and 11 differences between active inference and reinforcement learning one could probably quibble or update or lump or split but overall it did really well so I wanted to turn to the ant colony setting given all the above and alluding to contemporary discourse on applied Act of inference can you write me 23 specific situations where testable hypotheses arise from bare observation of ant colony behavior and 23 situations were provided different situations that ant colonies engage in trail formation division of labor brood sorting so on I wrote Let's delve into idea 7 nest building this concept is very interesting to me and a few of my nestmates as well can you do literature meta analysis into ant nest building and report back with an elegant blend of the most cited papers and the most relevant ones papers were returned 10 papers were returned I don't know which ones are highly cited and which ones are relevant there were some cut off answers but broadly speaking this looks pretty relevant since it's dealing with empirical and review work on ant nests so I wrote great can you write out the comprehensive pseudo code for starting out an agent-based active inference model which captures most of the relevant terminology from all the ant papers above and it did this the ant Behavior studies parameters were initialized the ant agent was defined in terms of its perception of the environment and action selection sense making and decision- making that is based upon minimization of free energy of course it's not fully written out here in the pseudo code but that's something that one could go further into a discussion with model or use standardized methods action performance and communication initialization running the simulation updating the environment optional introduction of environmental changes and maybe a finalized simulation routine that summarizes or gives visuals excellent I'm excited about this pseudo code which I summarized as an active inference model of within within and among task switching extra word there as being like the semantic konel or strange attractor of my PhD I'm going to provide you my first ideas for this seven chapter dissertation I'm thinking of after describing this sketch to you and absolutely including all of the above discussion we've had please comprehensively expand this into a table where the columns are chapter number chapter name chapter goal and chapter methodology first chapter more about literature review use social task allocation and active inference second chapter proposing mathematical framework for infra ANS task allocation third chapter developing a more physiological grounded extension for future integration with multiomic data fourth chapter is a collaboration with friend Who develops secure autonomous video deployments where we create a cheap and eco-friendly neuromorphic ant colony Behavior recording device that fits inside of a small backpack fifth chapter explicitly tackles the metaphysical and onto linguistic dimensions of the project sixth chapter describes the synthetic intelligence system enabling all the chapters in terms of requirements engineering seventh and final chapter encapsulates the entire dissertation with several important discussions connections and insights otherwise inappropriate for the six previous chapters and it provides those chapters six of them in a table I wrote wonderful please improve this table so that the existing insights are crystallized any missing relevant features are added and also add a new column for chapter relevance as distinctly articulated from chapter goal the chapter relevance column was provided and it does seem to differ from the goal and now just to close out this conversation and video I wrote last question for now can you come up with 27 hilarious possible titles for the overall project ideally including some acronym which spells out an ant related pun from the full technical title go wild thank you and I'll just read out the puns Antics btics antidote antsy antenna neuro Antics I ant intelligent Ant Farm analytic Quant an sin antic infra ant anticipate mbul an ant Venture cogn ant wander ant Dina ant Converse an frantic benevolent elegant Fon an consultant omnis an demsin ant ramp ant these titles blend humor with technical aspects of your project incorporating ant related puns and acronyms that reflect the essence of your research in a playful manner so this is just a first pass on the research GPT tool looks like great work by the SI base team and just very interesting and exciting as we all head into this scientific environment together all right see you around the Institute bye